

OpenDC-LCA: from heterogeneous evidence to bounded cooling decisions

One calculation spine keeps the functional unit, boundary, transformation, uncertainty and claim status visible.

DECISION QUESTION

Which cooling architecture is preferred, where can the ranking change, and what evidence would resolve it?

1 Evidence with declared role

Released lifecycle model

Microsoft/WSP component results
Role: arithmetic reconstruction

Public scenario evidence

eGRID · Boavizta · ÖKOBAUDAT
Role: bounded screening stress

Exchange-ready inventories

USLCI · GLAD · openLCA/Brightway
Role: registered, not silently combined

Primary engineering evidence

Performance maps · BOM · failures
Role: required for decision-grade claims

2 Harmonization and claim gate

1

Define equivalence

Useful computation · lifetime · reliability

2

Align what can be aligned

Unit · GWP basis · year · geography

3

Expose transformation limits

Interpolation · extrapolation · boundary gap

4

Propagate and classify evidence

Joint stress · provenance · review status

SCREENING ≠ COMPARATIVE ASSERTION

3 Decision outputs

Reproducibility

24 / 24 released totals reconciled
source IDs + equations + digests

Conditional decision boundary

96.7 kg CO₂e/MWh
cold plate / one-phase crossover

Robustness and next evidence

29.8% outside released anchors
99→55%^P first-rank stress frequency
Output: crossover + stress envelope + acquisition plan

● reconstructed ● screening transformation ● registered / unlinked ● primary / decision-grade

Every numerical claim retains its evidence class.